uc3m
AIRBUS

Airframe Technology Master Design
Lecture 01.
Introduce general objectives of Structures Design

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7 May 2025

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Concurrent Engineering.

Structural Design Mission

The mission of Structural Design Organization is to integrate the Airframe technologies needed to define aircraft structures to be manufactured, certified and maintained in service.

— Design, Develop, Qualify and Certify air vehicle structures

— Ensure Production, In-Service and continuous airworthiness support

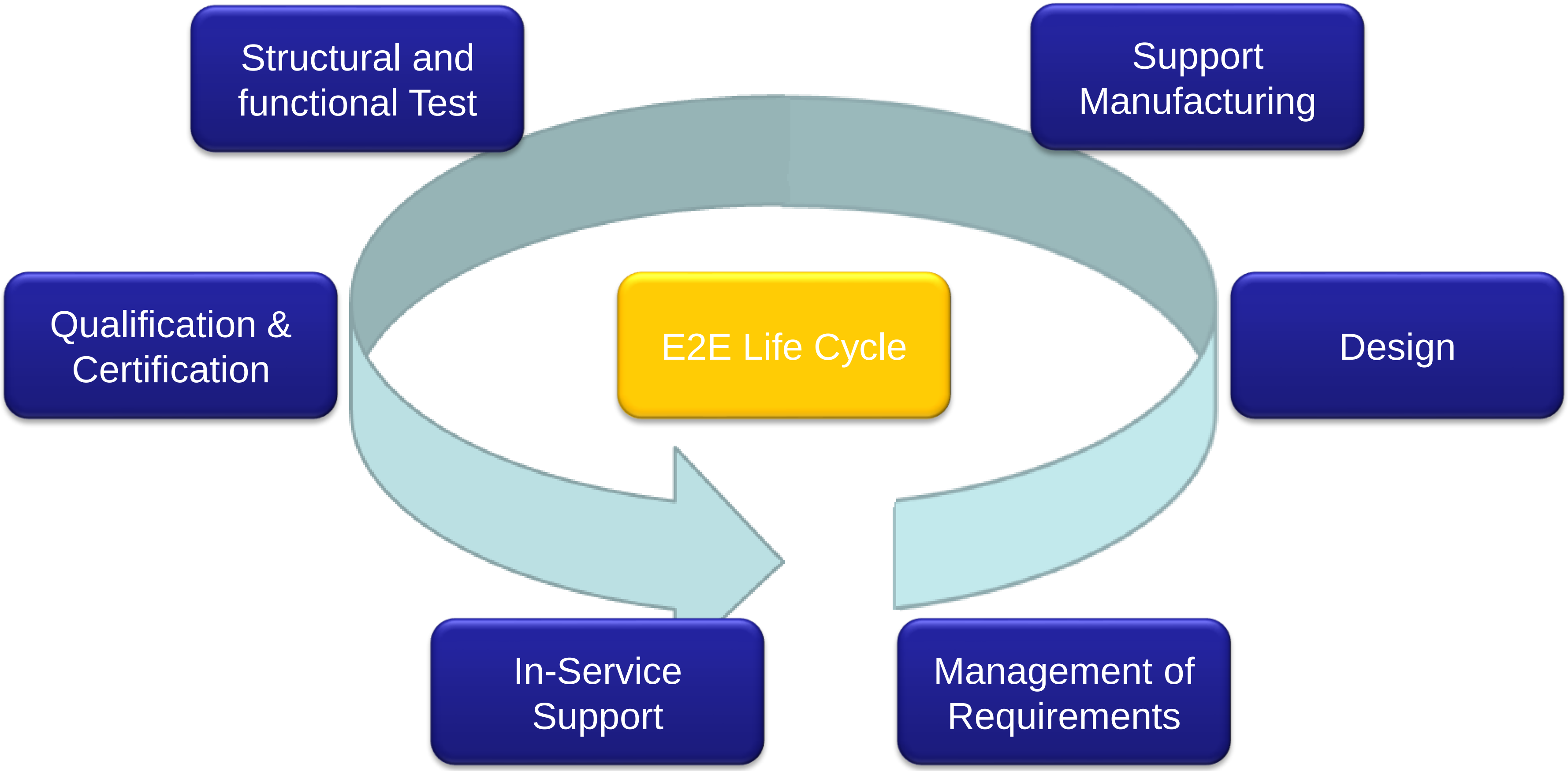
— Ensure support during full E2E life cycle

— Develop new technologies and methodologies for innovative and competitive air systems solutions

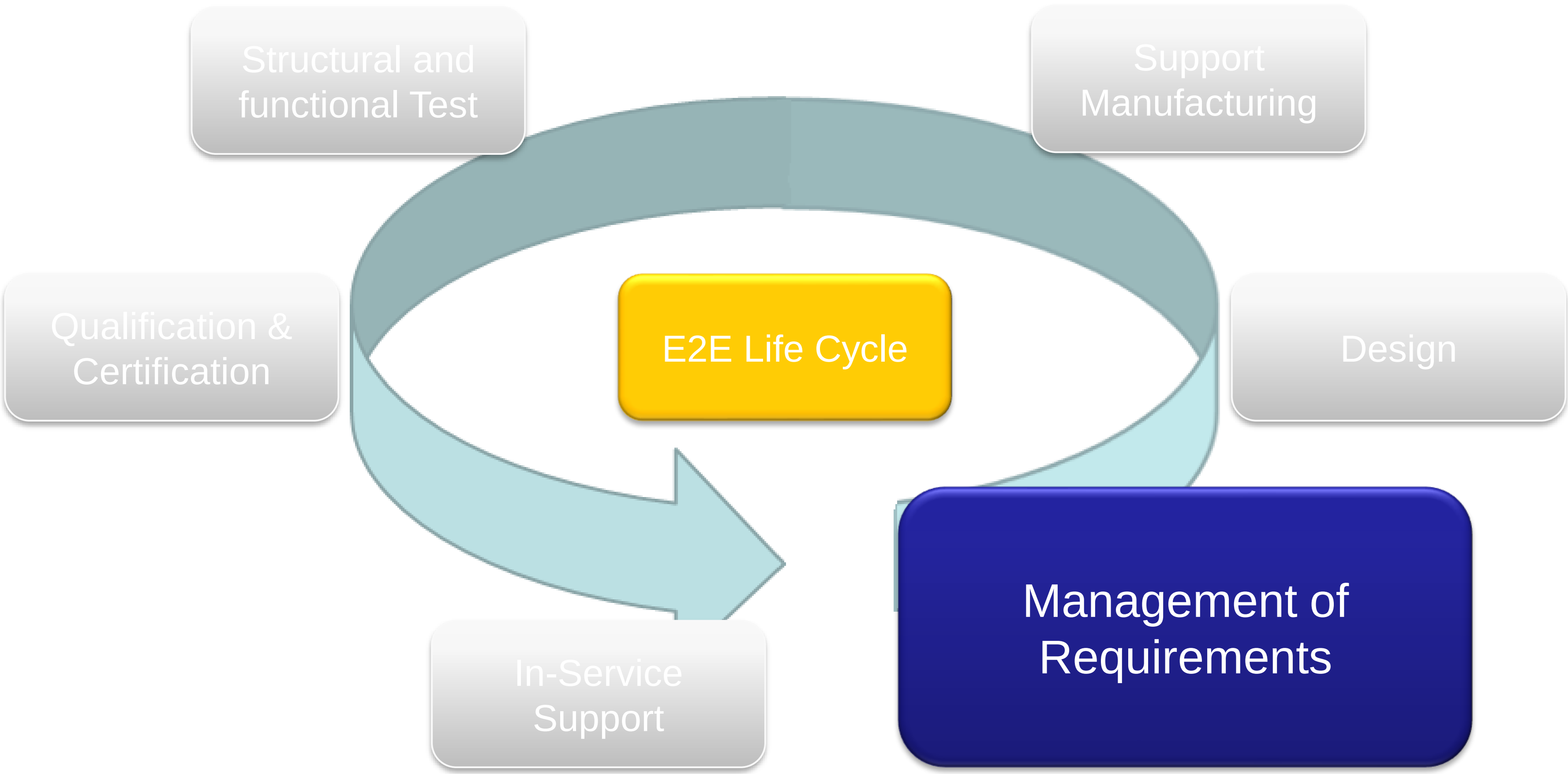
Typical organization is a matrix structure, with technical competences along one axis and programs along the other axis.

E2E life cycle

The product lifecycle describes all stages of the product throughout its life, starting from the expression of its need until the delivery and in-service support.



E2E life cycle



Definitions

- Requirement
  - is a formalized statement identifying a capability, a functionality, a physical characteristic or a quality that must be met or possessed by a system or system component to satisfy a contract, standard, a specification or other formally imposed documents (may be developed at any point in the product lifecycle).

- Requirements Management
  - concerns the collection, analysis, validation and verification of requirements with the purpose of defining and delivering the right product.

- Requirement Validation
  - is a set of activities to ensure that requirements are correct and complete so that the product will meet upper level requirements and user needs.

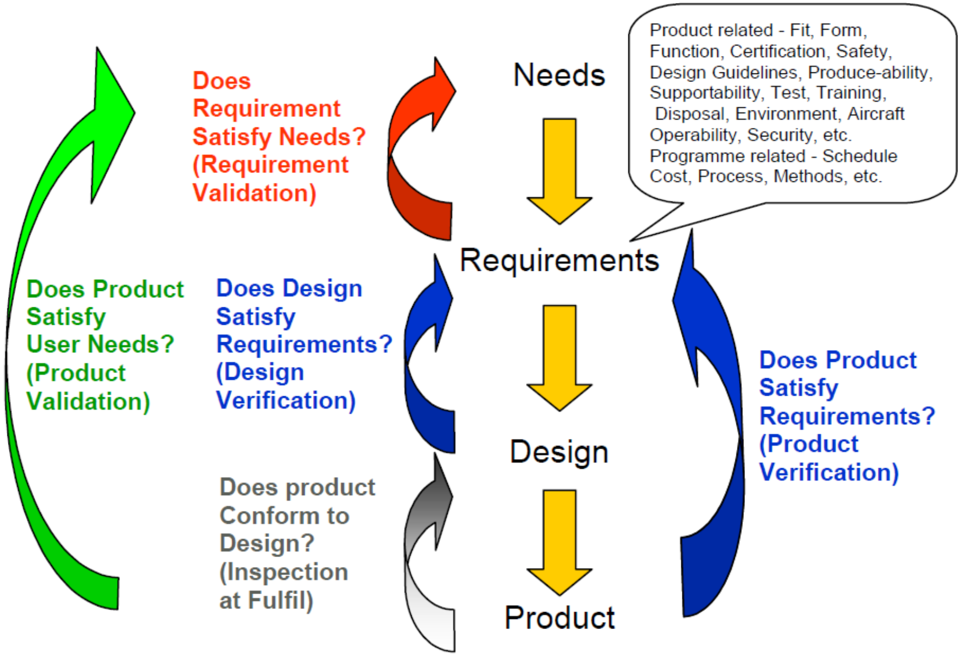
Definitions

— Design Verification

- is the evaluation of implementation of design (architecture, components…) against requirements to determine that they can be met.

— Product Verification

- is the evaluation of implementation of product against requirements to determine that they have been met.



Basic Engineering Process

Management of Requirements

MoC codes

Compliance task	Means of Compliance	Associated Compliance Documents
Engineering Evaluation	MC0: Compliance statement • Reference to Type Design documents • Election of methods, factors, etc. • Definitions	Type Design Documents, Recorded Statements
	MC1: Design Review	Description, Drawings
	MC2: Calculation/Analysis	Substantiation Reports
	MC3: Safety Assessment	System Safety Analysis
Tests	MC4: Laboratory Tests MC5: Ground Tests on A/C MC6: Flight Tests MC8: Simulation	Test Programs, Test Reports, Test Interpretations Note 3 
Inspection	MC7: Inspection (on Mock-up, A/C, etc.)	Reports of Inspection Visits
Equipment qualification	MC9: Equipment Qualification	See Note 1

Note 1: Equipment Qualification is a process, which may include all previous means of compliance.

Definitions

Equipment Qualification

Qualification is the set of tasks contributing to provide proofs, that the defined product satisfies the specified need and can be produced. The customer attests that the defined product, identified by the design data file, meets all the requirements of the **Technical specification** and can be produced.

Aircraft Certification

Certification is the process demonstrating that the Design of an Aircraft complies with the applicable aviation requirement

Main activities in Structural Design

- Perform validation and verification activities at equipment, component and aircraft level for qualification and certification requirements.
  - activities to ensure that requirements are correct and identify a MoC´s so that can be verified.
- Analyze and define the MoC’s for certification req’s to obtain the certification of the product.
  - concerns the collection, analysis and verification of certification requirements with the purpose of Designing a Certifiable structure.
- Monitoring, evaluation and analysis of development and laboratory test results.
  - activities to demonstrate that requirements are meet (MoC´s 4) so that the product will meet upper level requirements.
- Preparation and Approval of the Equipment Technical Specification and control of the supplier qualification program.
  - concerns the collection of qualification requirements with the purpose of Defining the PTS.

Link to Practical Exercise

Structure Requirements: Structural Integrity

- Rigidity: structural components must ensure that the outer surfaces are within the limits established.

- Strength requirements are specified in terms of:
  - limit loads (the maximum loads to be expected in service)
  - and ultimate loads (limit loads multiplied by prescribed factors of safety). Unless otherwise provided, prescribed loads are limit loads.

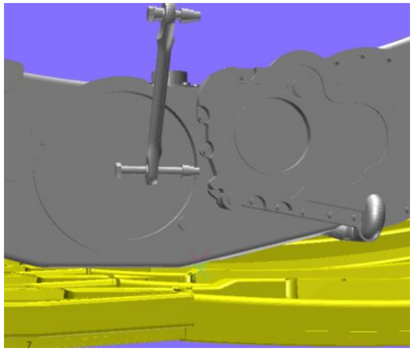
According to CS 25.305 Strength and deformation

(a) The structure must be able to support limit loads without detrimental permanent deformation.
At any load up to limit loads, the deformation may not interfere with safe operation.

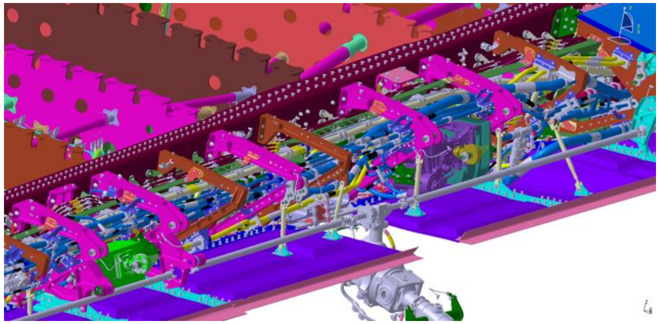
(b) The structure must be able to support ultimate loads without failure for at least 3 seconds.

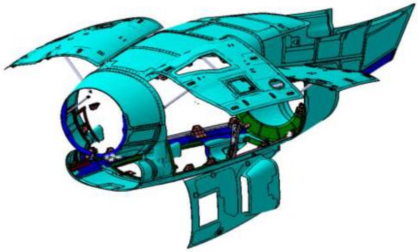
Structure Requirements: Functional and operational

- Overall geometry
- Available space
- Bolt installation

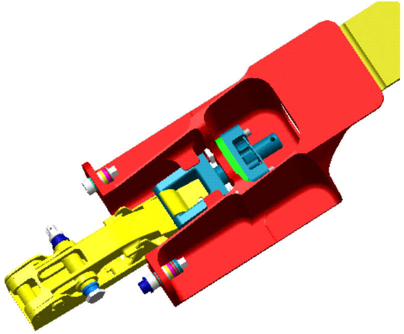


- Interchangeability, for example the cabin floor panels
- Inspection
- Maintenance/reparability
- System installation





- Interface constraints, like room available for the component
- Aerodynamic

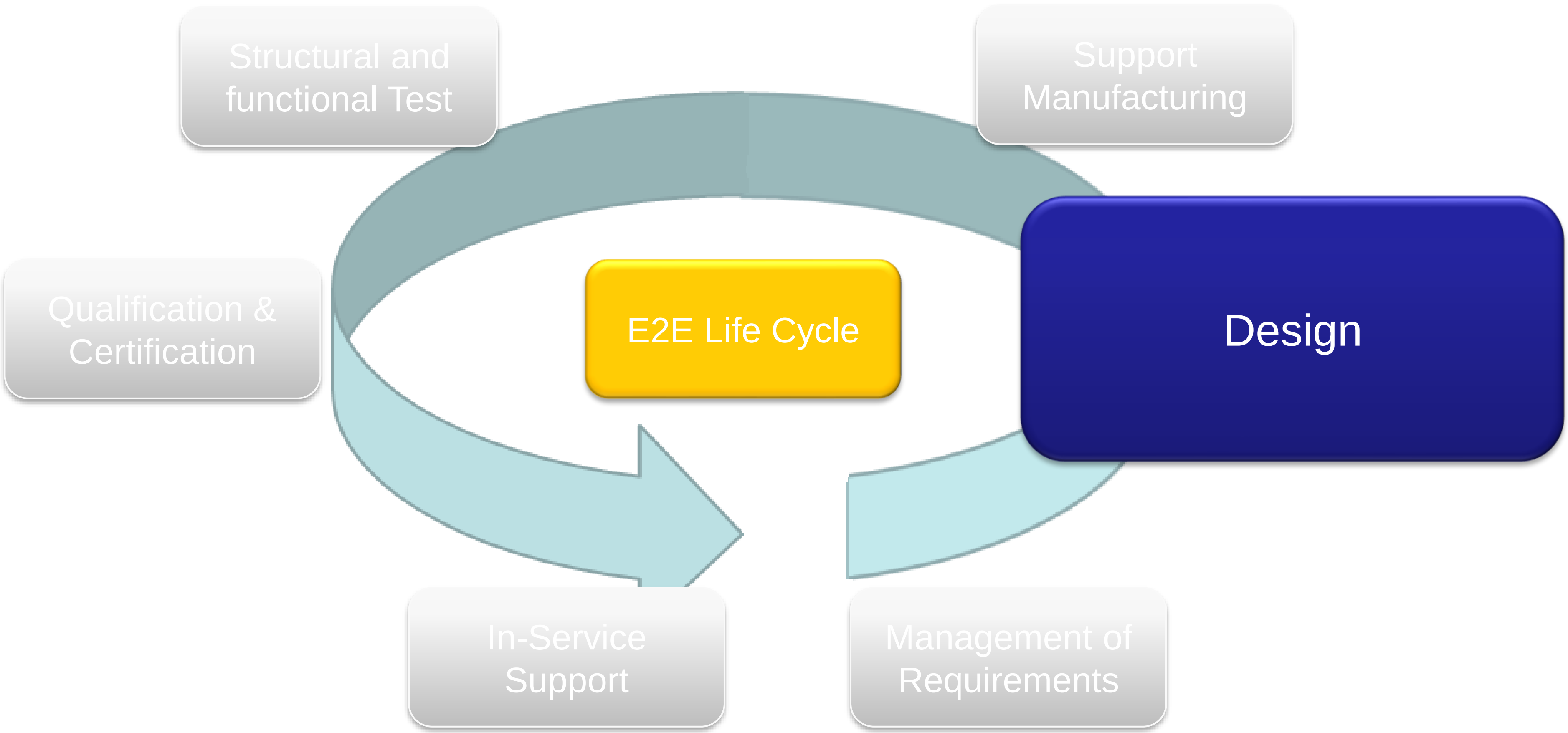


Structure Requirements: weight

Weight saving is one of the main drivers for any structure Design.

- A trade off study should be performed to compare the different concept according with the available technology.
- The design of the structures should be optimized toward a minimum installation of elements or riveting. For this purpose high integrated structures should be considered, ie use of composite material or machined vs sheet metal

E2E life cycle



Functions and responsibilities

The Structural Design department Functions and responsibilities are:

- Study of the different alternatives of Structural design concept in relation with the manufacturing processes and assembly sequences to fulfil the requirements.
- Define in conjunction with manufacturing engineering (concurrent engineering) the technologies of manufacturing to be applied on each component.
- Definition of the design principles.
- Establish Product Configuration.
- Materials selection, standards processes and protections considering the environmental impact and applicable regulations.
- Release of the relevant design documents dossier to support the different program milestones in time and cost.
- Design dossier documentation (drawings, 3D files, DMU generation and update, BOM and technological documents), checking and approval.

Functions and responsibilities

The Structural Design department Functions and responsibilities:

- Structural parts and assemblies weight data definition and design to achieve weight targets.
- Support the Design Reviews for the different phases of the programs according to the specific schedules and milestones.
- Provide support for the new offers definition.
- Agree system installation integration.
- Stablish configuration and CAD methods and tools to apply during complete development.
- Definition of development tests and certification test establishing the associated documentation.
- Analyze and define the MoC’s for certification/qualification req’s to obtain the certification/qualification of the product.

Functions and responsibilities

The Structural Design department Functions and responsibilities:

- Provide support for manufacturing non-conformances
- Definition of in-service repairs including AOG situation.
- Design approval for Concessions during the production development.
- Analysis of the Airworthiness Directives (AD) and SSBB of the in-service fleet including safety recommendations and actions.
- Support the in-service failures and support to Product Safety Domain for in-service events including incidents and accidents.

Structural Design role

Competences

The Design role includes a set of competencies depending on the associated technology, main of them can be summarized as follows:

Design Metal & Composite Components

- Metallic structure
- Composite structure
- DMU - Digital Mock-up
- CAD Model
- Surface Protection
- Materials & Processes
- Requirements management
- Interfaces management
- Tolerances analysis
- Qualification & Certification
- Design to cost
- Structural Test

Modifications & Repairs during hole life cycle

- Non conformity process: ERC, DQN, HNC and Concessions (manufacturing and assembly)
- Modification process
- Answer Customer Technical Queries

AM Part

Definition of Design principles

- Design principles Metal
- Design principles composite
- Design Manual
- Check List

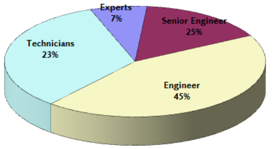
Machined Part

Sheet Metal Part

esign Mechanisms

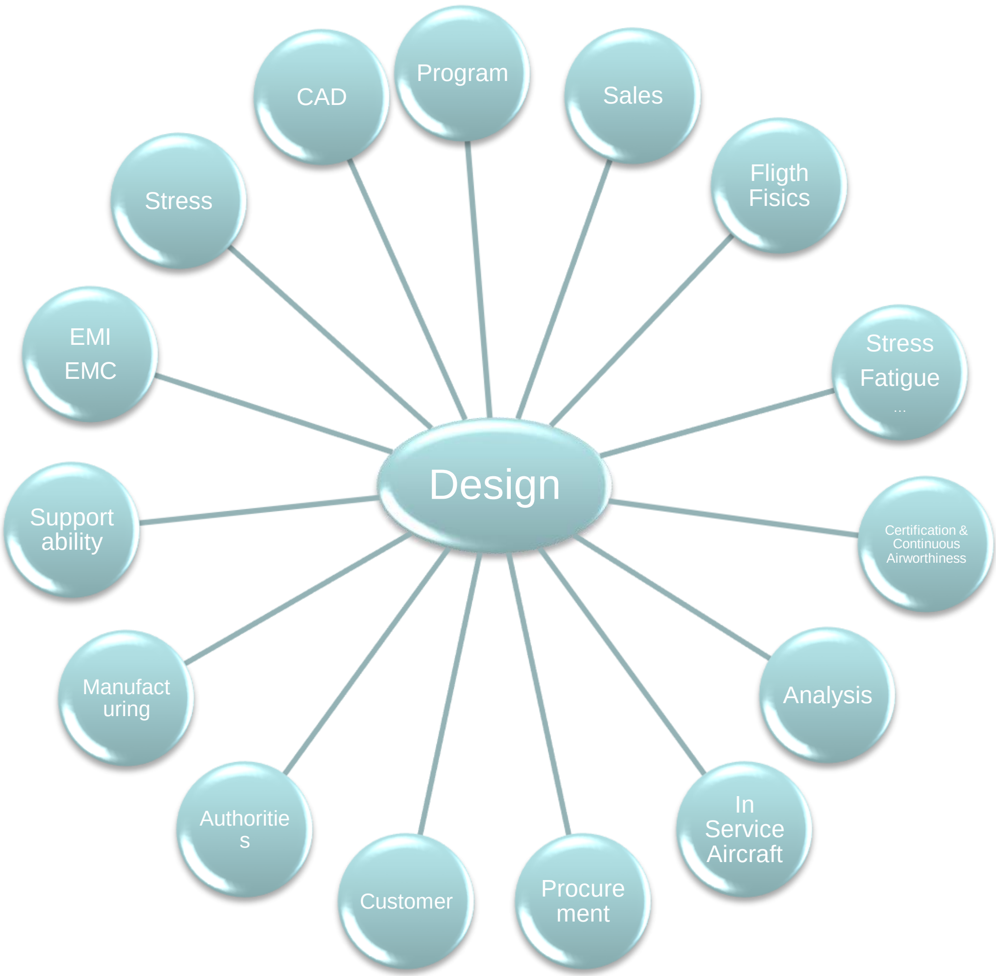
- Doors and Mechanisms
- Landing gear and Mechanism
- Flight control

Flap Rollers Mechanism



Interfaces and Stakeholders

The Structural Design department work in a continuous concurrence with:



Design tools

Tools for product development

Product design development CAD/PDM tools customised to support Concurrent Engineering model with Full 3D methodology .

- Design tools: CATIA V4 & CATIA V5
- Single Vault with real time access to CAD and Product structure from all Groups.

Product structure:

- SPRINT.NET/CMS
- PRIMES
- OPTEGRA/CMS

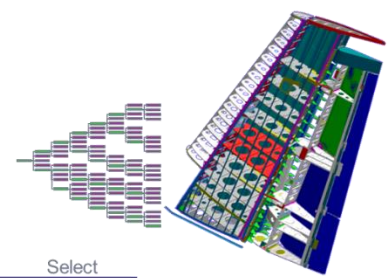
Visualizers:

- Product View

Management of approval flows.

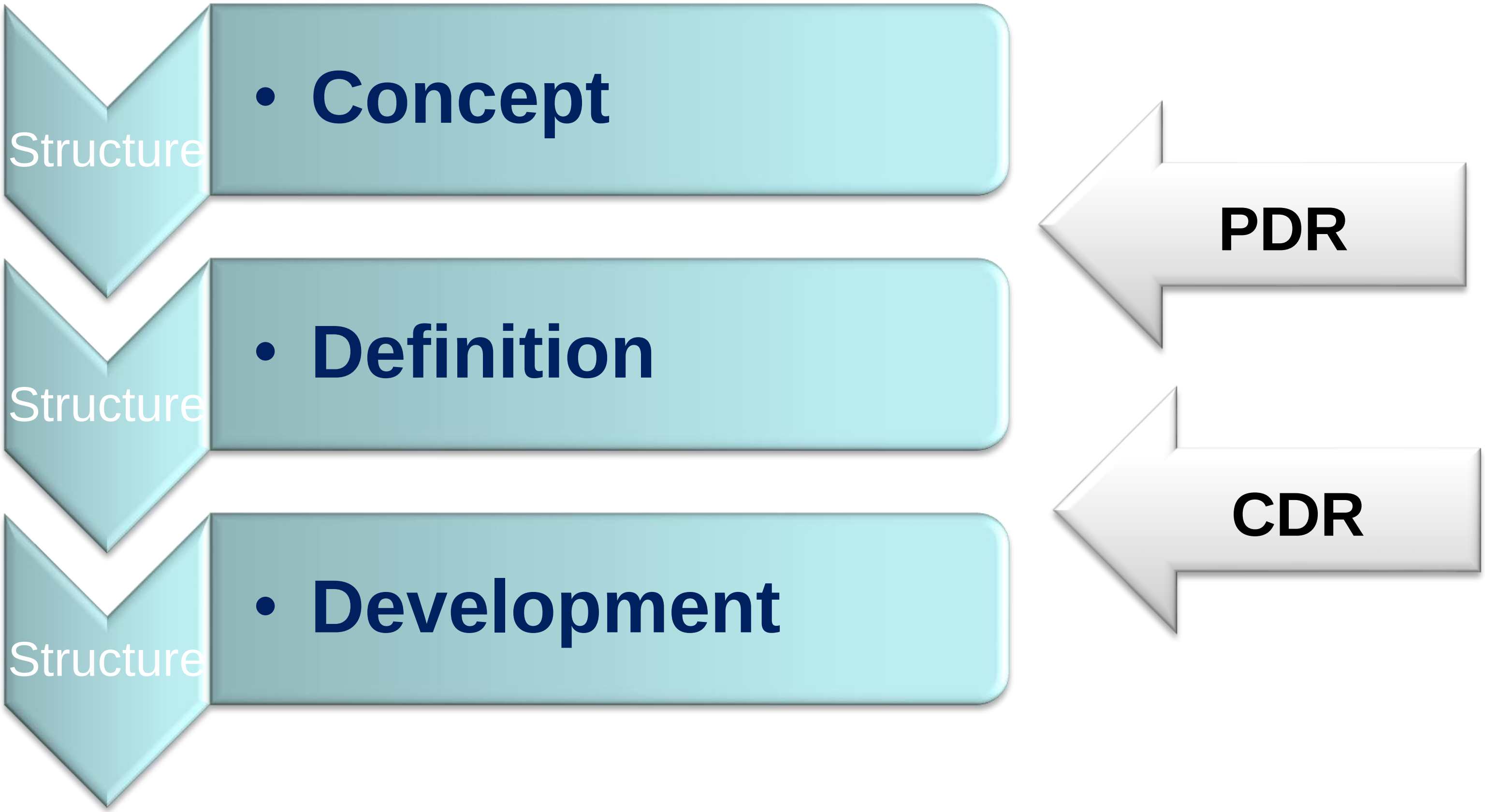
Knowledge Management tool, with access to:

- Design manual
- Design principles
- Check lists
- Design academy
- Lessons learnt



Process phases

The design process is divided in 3 main phases:



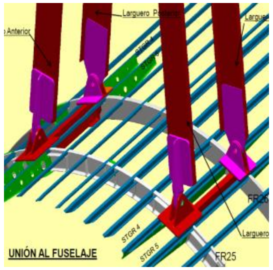
Process phases

Design Concept

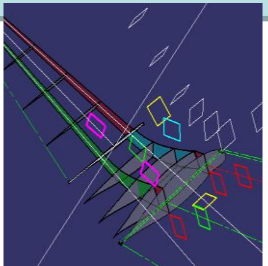
Conceptual Design:

<u>Inputs</u>
- **A/C Global Master Geometry**
- **Flight control surfaces**
- **Technical specification (top level requirements)**
- **Contractual requirements**
- **Material data**
- **Certification rules**
- **Manufacturing requirements**
- **Systems preliminary requirements**
- **Load estimation**

<u>Activities</u>
- **Analysis of Requirements.**
- **Check that the Aircraft Concepts in the Initial Product Configuration (definition) are feasible from structural point of view**
- **Trade-offs , studies , Lay-outs**
- **Verify that systems , equipment and routings have the necessary space allocation**
- **Preliminary weight estimations**
- **Selection of materials**
- **Interchangeability criteria**
- **Design Principles and applicable rules**
- **Engineering concurrency**
- **Planning & Control**



<u>Outputs</u>
- **Validated structural concept and alternatives selection:**
  - **Establishment of the design concept for every element as result of the concurrent engineering process.**
- **Space Allocation Mock-up**
- **Preliminary design principles**
- **Identification and initial definition of the different interfaces with surrounding structure and systems.**
- **Initial sizing**
- **Preliminary stiffness requirements, mass distribution and dynamic loads**
- **Target weights for components**
- **Applicable rules and processes**
- **Master Schedule**



Process phases

Design Definition

Definition:

Inputs

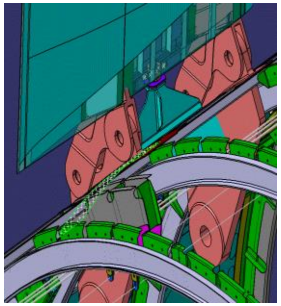
- Outputs from structure concept
- Technical specification and contractual requirements.
- Top Level A/C requirement
- Updated Global Master Geometry , Surfaces and Material data
- Certification rules
- Preliminary Load
- System installation requirements
- Supportability requirements
- Aerodynamic requirements
- Preliminary loads
- Preliminary Acoustic and Vibration spectrum

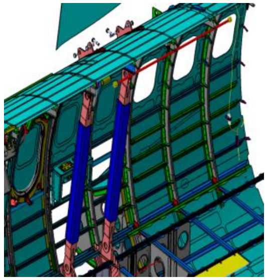
Outputs

- Certification plan
- Test plan
- FEM
- Sizing and structural analysis documents (internal loads, allowable, stiffness…)
- 3D model for each component.
- Product structure
- Lists of material and STD
- Weight targets for each component detailed distribution
- Stiffness, mass distribution and updated dynamic loads
- Equipment specification

Activities

- Definition of Design Solutions
- DMU update
- Product structure definition
- Interfaces and frontiers definition
- Materials and standards selection
- Engineering Concurrency
- Configuration
- Specification for components
- Support to Manufacturing





Process phases

Design Development

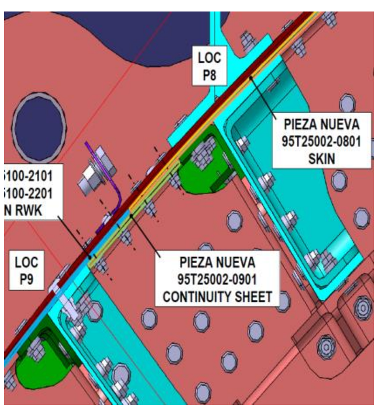
Development:

Inputs

- Outputs del structure definition
- Updated A/C Global Master Geometry and Surfaces
- Material data
- Systems requirements
- Manufacturing requirements
- Supportability requirements
- Certification rules
- Design Loads
- Certification Load

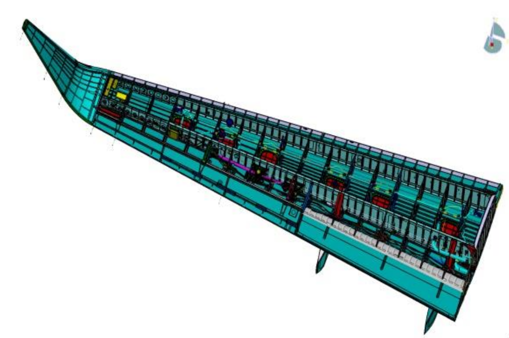
Outputs

- 3D model for each component.
- PLY model
- DRW (if applicable)
- Product structure
- List of material and std
- Weight for each component
- Equipment specification
- Test results
- Structural Analysis documents



Activities

- Detail design solutions and product configuration
- Test
- Manufacturing & Assembly Tolerances
- Design Documentation Release
  - Drawings
  - Part Lists
- Drawing/Models Checking List
- Modification Follow-up



Design Reviews

Definition and Objectives:

- A Design Review is a formal, documented, systematic and controlled review of the design achievements at planned intervals throughout the whole product development process to verify the conformity to the appropriate and agreed requirements, e.g. Top Level Aircraft Requirements (TLAR).

- Design Reviews (PDR/CDR) are performed for steering and monitoring the activities needed to deliver on time, on quality and as per customer expectation.

- A Design Review ensures that all design goals relating to the multidisciplinary aspects to be reached by that point are examined with achievements recorded, problem areas and inadequacies identified with corrective actions initiated and follow-up review planned. The results of the review enable the decision for the program to proceed to the next phase of the plan at some level of managed risk.

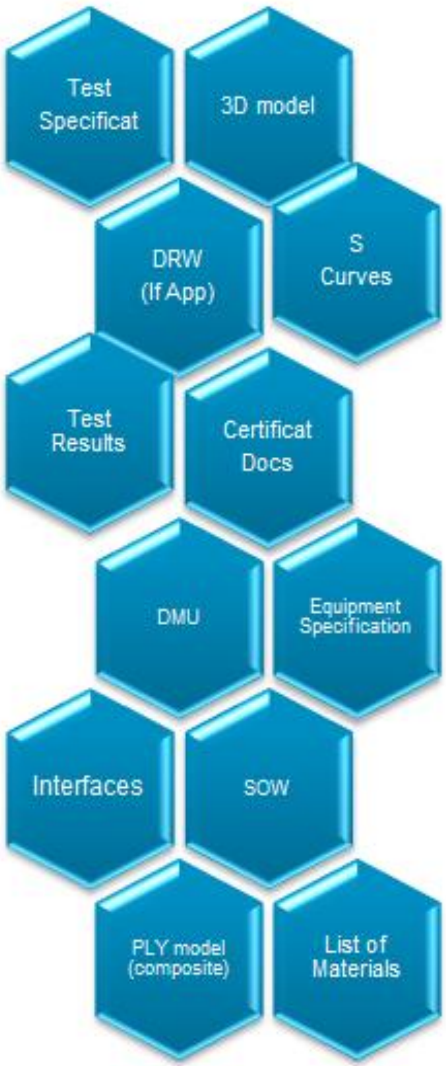
Design Reviews

Typical agenda for Design Review:

- Component Description
- Aerodynamics / loads
- Thermal
- Structural justification
- Failure mode & effect analysis
- System
- Lightning protection and electrical bonding
- Certification
- Weight
- Production
- Maintainability program
- Compliance to the specification
- Conclusion - Actions summary

Link to Practical Exercise

Deliverables

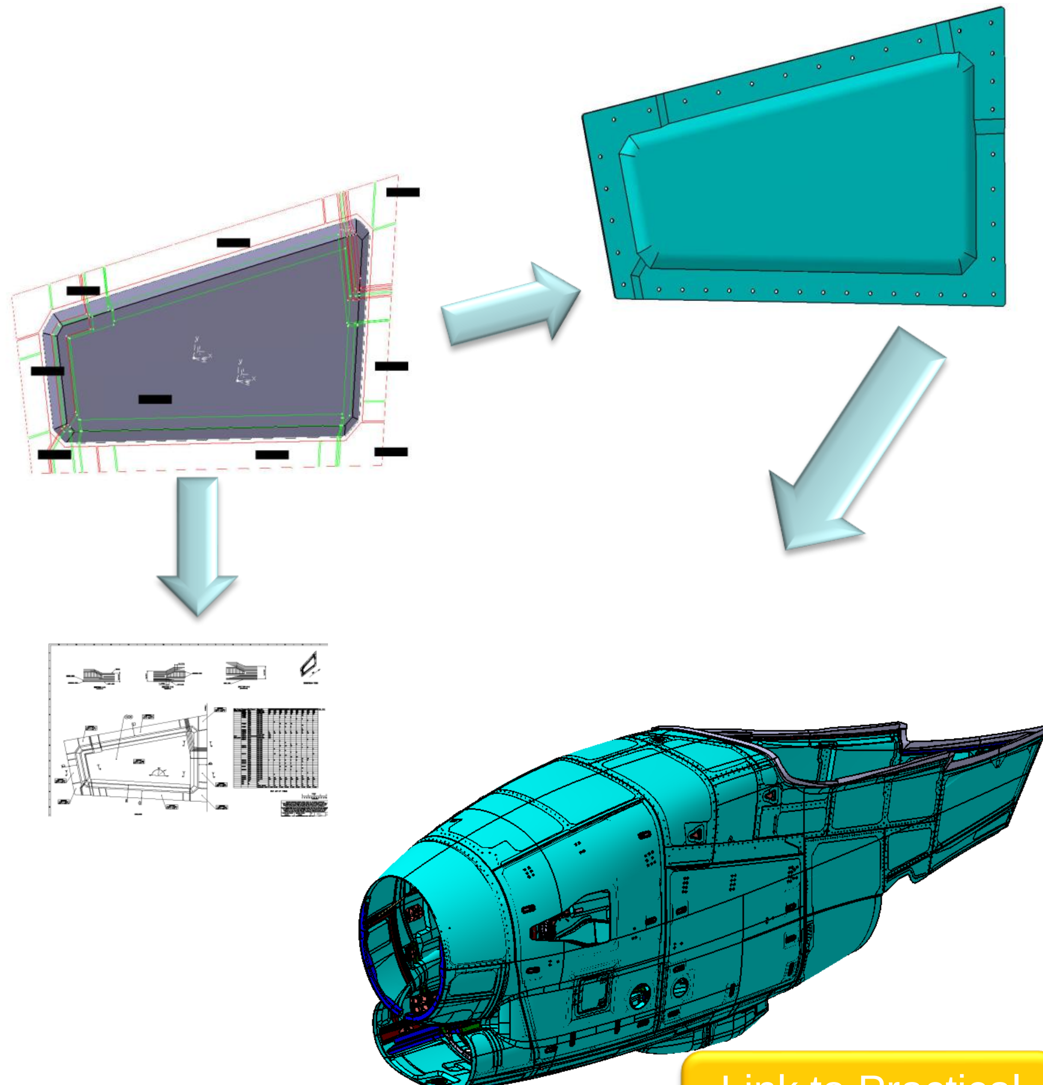


Deliverables

Drawing set

Design deliverables:

- 3D model for each component
- DMU
- PLY model (composite)
- DRW (if applicable)
- Product structure
- List of Material and STD
- Equipment specification
- Test results



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Bill Of Material Schedule — A/C PROGRAMME

LOC	DRAWING SET NUMBER	ISS	STATE	DRAWING SET TITLE	DESIGN OFFICE CODE : MG	PROT TREAT CHART	PROJECT TEAM
00		B00	RELEASED	PANEL ASSY-LOWER REAR		TNA00710003	MPNC

GENERAL INFORMATION

LO/ALT		QTY 003	QTY 002	QTY 001	QTY 000	UNIT	ZONE	ITEM	PART NUMBER	ISS	DESCRIPTION SPECIFICATION	OH	SP	PROTECT TREAT	I D	ICY	REMARKS
					X		01A09		000	A00	PANEL ASSY-LOWER			000	N/A	NA	
				X			01A09		001	A00	PANEL ASSY-LOWER	000		000	N/A	NA	
			X				03B11		002	B00	PANEL ASSY-LOWER			000	N/A	NA	010
		X					03B11		003	B00	PANEL ASSY-LOWER	002		000	N/A	NA	010
					1.0	Each	02D06	0	200	A00	CORE			000	N/A	NA	
				1.0		Each	02D06	1	201	A00	CORE	200		000	N/A	NA	
		1.5	1.5	1.2	1.2	m2		2	Z19776		CARBON PREPREG						
		0.2	0.2	0.2	0.2	m2		3	ABS5319AA1		ADHESIVE FILM						
		0.2	0.2	0.2	0.2	m2		4	ABS5317BA2		EPOXI PRPREG MESH						
		0.3	0.3	0.3	0.3	m2		5	ABS5319AB2		EPOXY ADHESIVE F						
		0.2	0.2	0.2	0.2	m2		6	Z14512		POLYVINIL FLUORI						
			1.0			Each	03G12	10	202	B00	CORE			000	N/A	NA	
		1.0				Each	03G12	11	203	B00	CORE	202		000	N/A	NA	

DRAWING SET REMARKS

001	ROUND OFF EDGES
002	MANUFACTURE PER AIPS03-07-007

Link to Practical Exercise

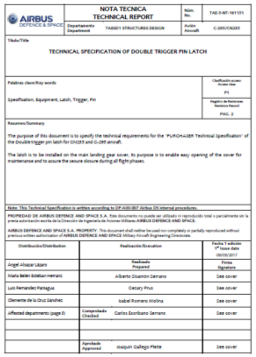
Product Technical Specification

Design deliverables:

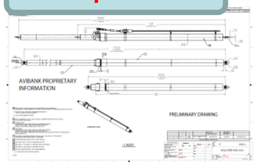
— Equipment specification:

- Preparation and Approval of the Technical Specification.
- Bidding Process.
- Offers evaluation and supplier selection.
- Support the delegation of signatures process of approval of the suppliers.
- Development Process and Validation of the supplier's documentation.
- Qualification of the Appliance(DDP)

Link to Practical Exercise


NOTA TECNICA
TECHNICAL REPORT
TECHNICAL SPECIFICATION OF DOUBLE TRIGGER PIN LATCH

Compl Matrix

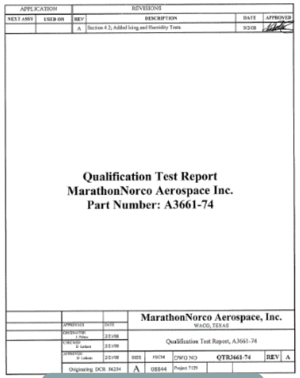



Qualification Test Procedure

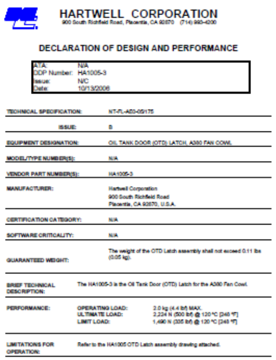
MarathonNorco Aerospace Inc.
Part Number: A3661-62

Pre-Release DRAFT

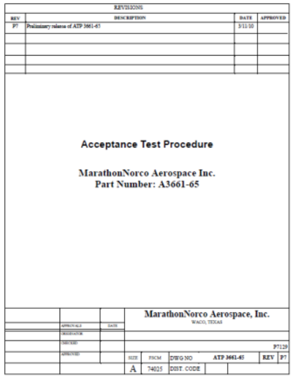
MarathonNorco Aerospace, Inc.


Qualification Test Report
MarathonNorco Aerospace Inc.
Part Number: A3661-74

MarathonNorco Aerospace, Inc.


HARTWELL CORPORATION

DECLARATION OF DESIGN AND PERFORMANCE


Acceptance Test Procedure

MarathonNorco Aerospace Inc.
Part Number: A3661-65

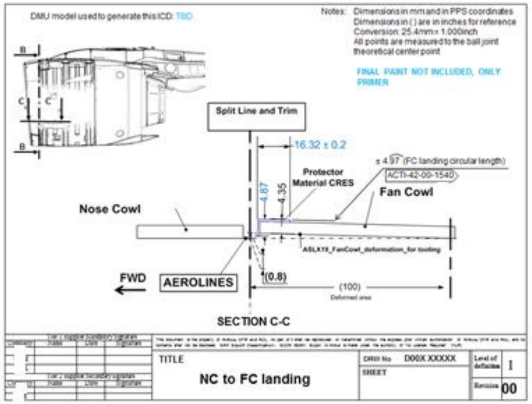
MarathonNorco Aerospace, Inc.

Technical Spec | DRW | QTP | QTR | DDP | ATP

ICD

Frontier and Interface Drawings

- Set of data used to validate information at a junction between different structural components, systems or equipment's ((typically parts developed in parallel by different suppliers (internal or external)).
    - They can include different kind of CAD models (3D models or 2D drawings) and all other associated documentation (Technical notes, FEM models, Sets of interface loads…).



ICD

Frontier and Interface Drawings

- ICD documents content
  - Views of the interface assembly (all parts)
  - Cross section showing the two mated parts with nominal geometric definition of the interface
  - Vectors visual definition to ensure accurate understanding of the interface orientation.
  - Geometrical tolerances (relative to dedicated datum system) and stack-up tolerances (geometrical, displacements, thermal…)
  - Functional clearances between both sides of the interface. (Hinges gaps, bumpers gaps, mounts links gaps… etc)
  - Free state seals or springs, functional crushing ranges (nominal, min and max) and crushing loads (load versus crushing).
  - Aerosmoothness singularities (deviations from lofted contour) internal to each component
  - Aerosmoothness steps and gaps ranges between all interfaced components
  - Materials and protections in contact at interface.
  - Standard items references (bolts, electrical and fluids connectors, etc)
  - DMU model number or drawing used for the generation of the data.

Link to Practical Exercise

Certification Documentation

ACD2 Structural Description

Núm. No. TAE-FF-DT-150027	**PORTADA DE FIRMAS** SIGNATURE COVERSHEET		Programa / Programme: FF - A330 MRTT - FRANCE
Versión/Issue: B	Clasificación/ Access Class: None	Departamento/Department: STRUCTURES DESIGN	ATA: 51 - STANDARD PRACTICES AND STRUCTURES,52 - DOORS,53 - FUSELAGE,55 - EMPENNAGE,57 - WINGS

Título:
Title:

ACD2: Structural Description

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Authors	Checkers	Approvers
Nombre/ Name: Nuñez Martinez, Carlos Alberto SIGLUM: TAESDT # STRUCTURES DESIGN Fecha de Firma/ Date of Signature: 02/06/2017	Nombre/ Name: Pozzi Villalobos, Carlos SIGLUM: TAESDT # STRUCTURES DESIGN Fecha de Firma/ Date of Signature: 02/06/2017	Nombre/ Name: Sanjuan Olmo, Juan Andres SIGLUM: TAESDT # STRUCTURES DESIGN Fecha de Firma/ Date of Signature: 06/06/2017
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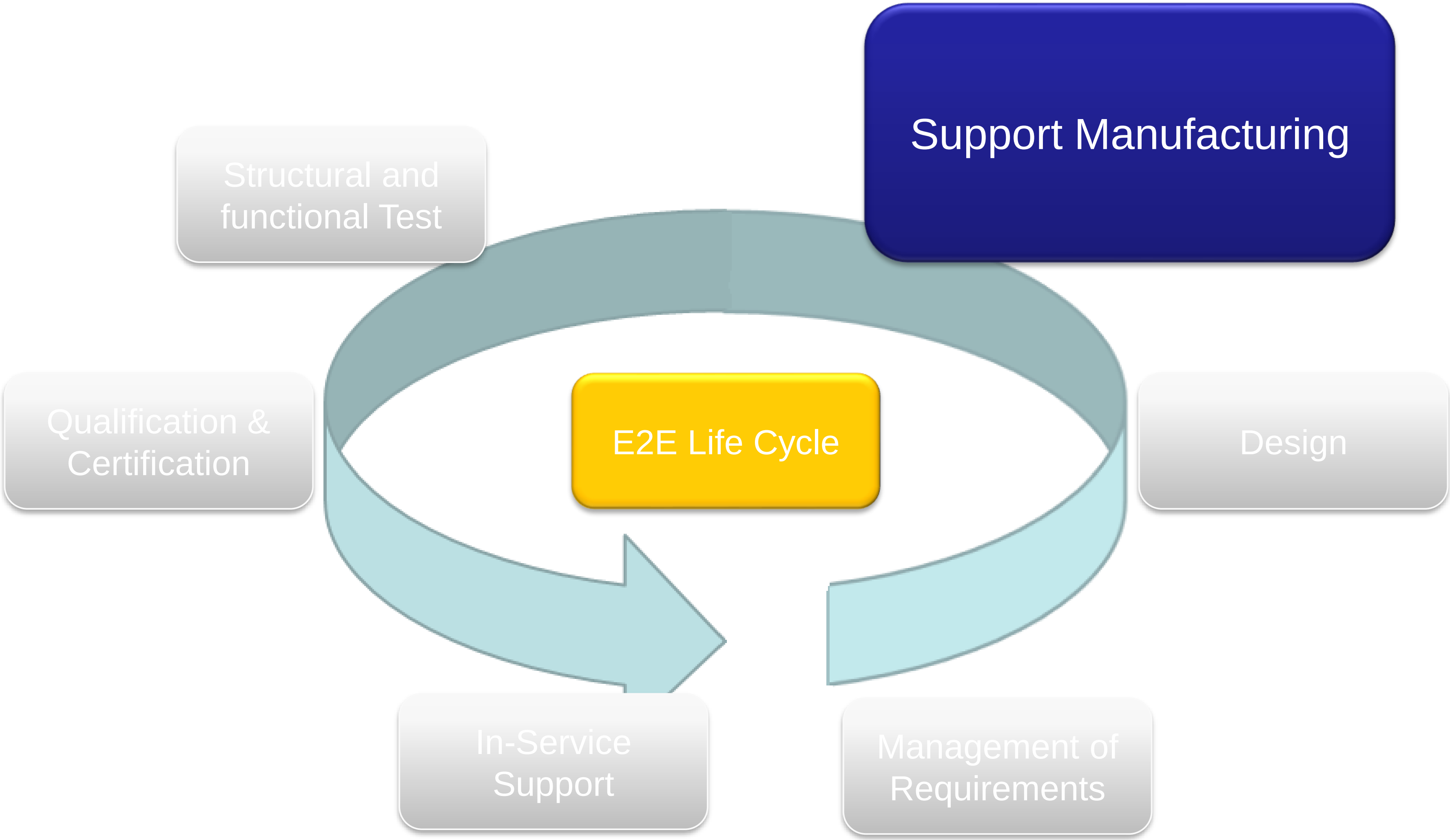
Nombre/ Name: N/A SIGLUM: N/A Fecha de Firma/ Date of Signature: N/A	Nombre/ Name: N/A SIGLUM: N/A Fecha de Firma/ Date of Signature: N/A	Nombre/ Name: N/A SIGLUM: N/A Fecha de Firma/ Date of Signature: N/A

Link to Practical Exercise

E2E life cycle

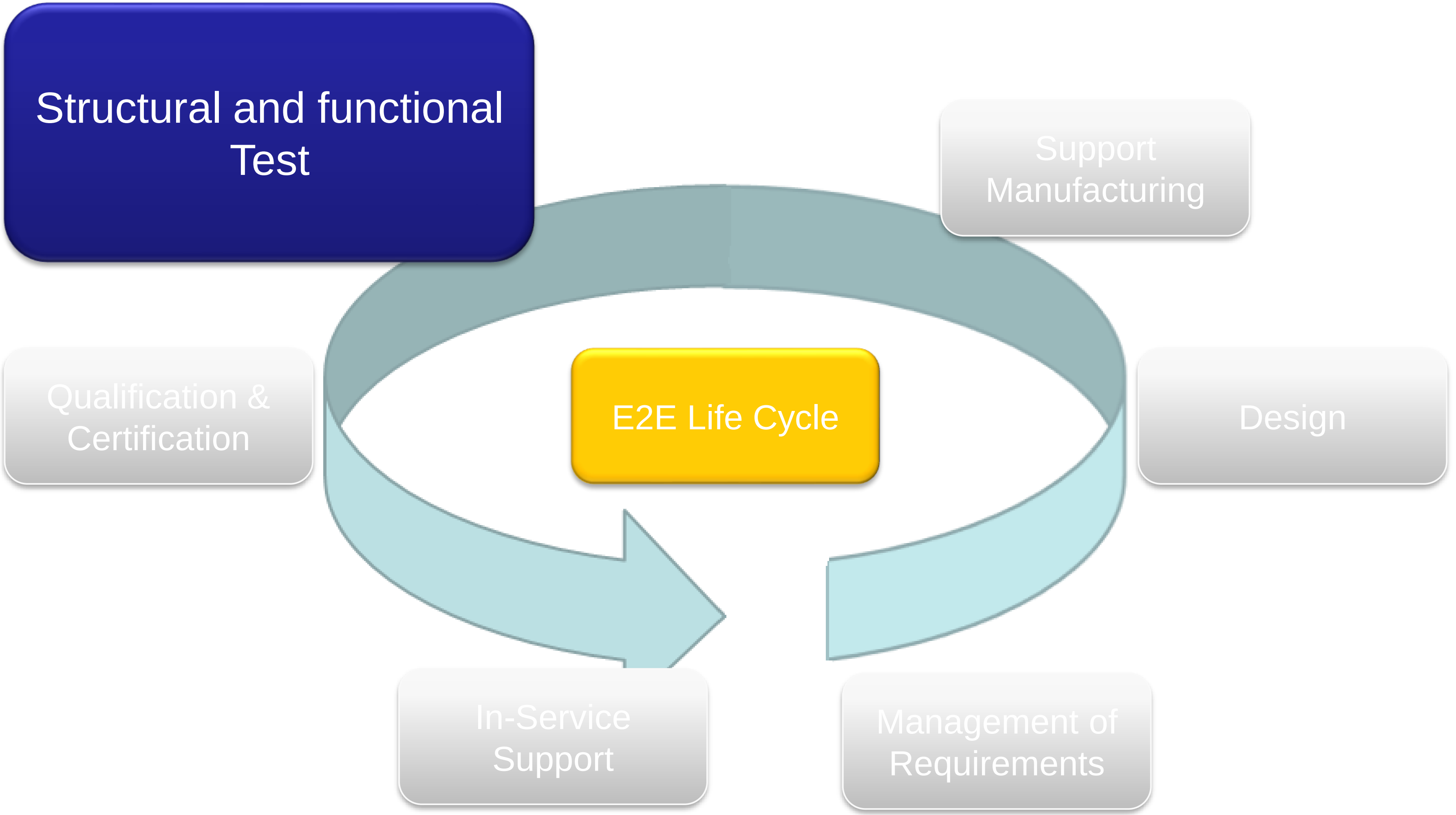


Main activities in Structural Design

- Develop innovative Metallic/Composite and/or Assembly technologies aligned with product strategy and ensure timely readiness for integration and operational availability.
- Preparation of the documentation for manufacturing and assembly.
- Resolution of nonconformities during manufacturing and assembly of parts and components.
- Evaluation of the Modification Proposals in co-operation with other technical areas.
- Preparation and approval of the technical solution of the Modification.
  - Planning and performing the needed actions to promote the Modifications to final Stage.

Link to Practical Exercise

E2E life cycle



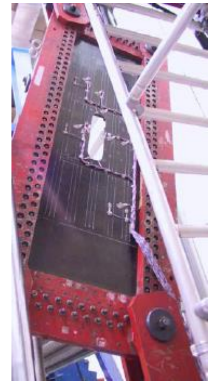
Main activities in Structural Design

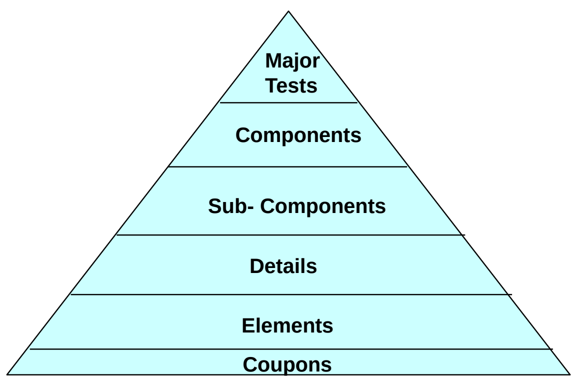
- Qualification/Certification test.
  - Preparation of the Test Program.
  - Preparation of the Test Plan.
  - Monitoring and execution of the Tests.
  - Contribution to definition of testing facilities (test rig)
  - Definition of components and requirements for testing
  - Analysis of the Test Results.

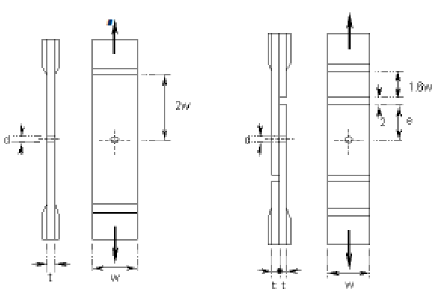
- Development Test.
  - Preparation of the Test Program.
  - Preparation of the Test Plan.
  - Monitoring and execution of the Tests.
  - Contribution to definition of testing facilities (test rig)
  - Definition of components and requirements for testing
  - Analysis of the Test Results.

Link to Practical Exercise

Main activities in Structural Design

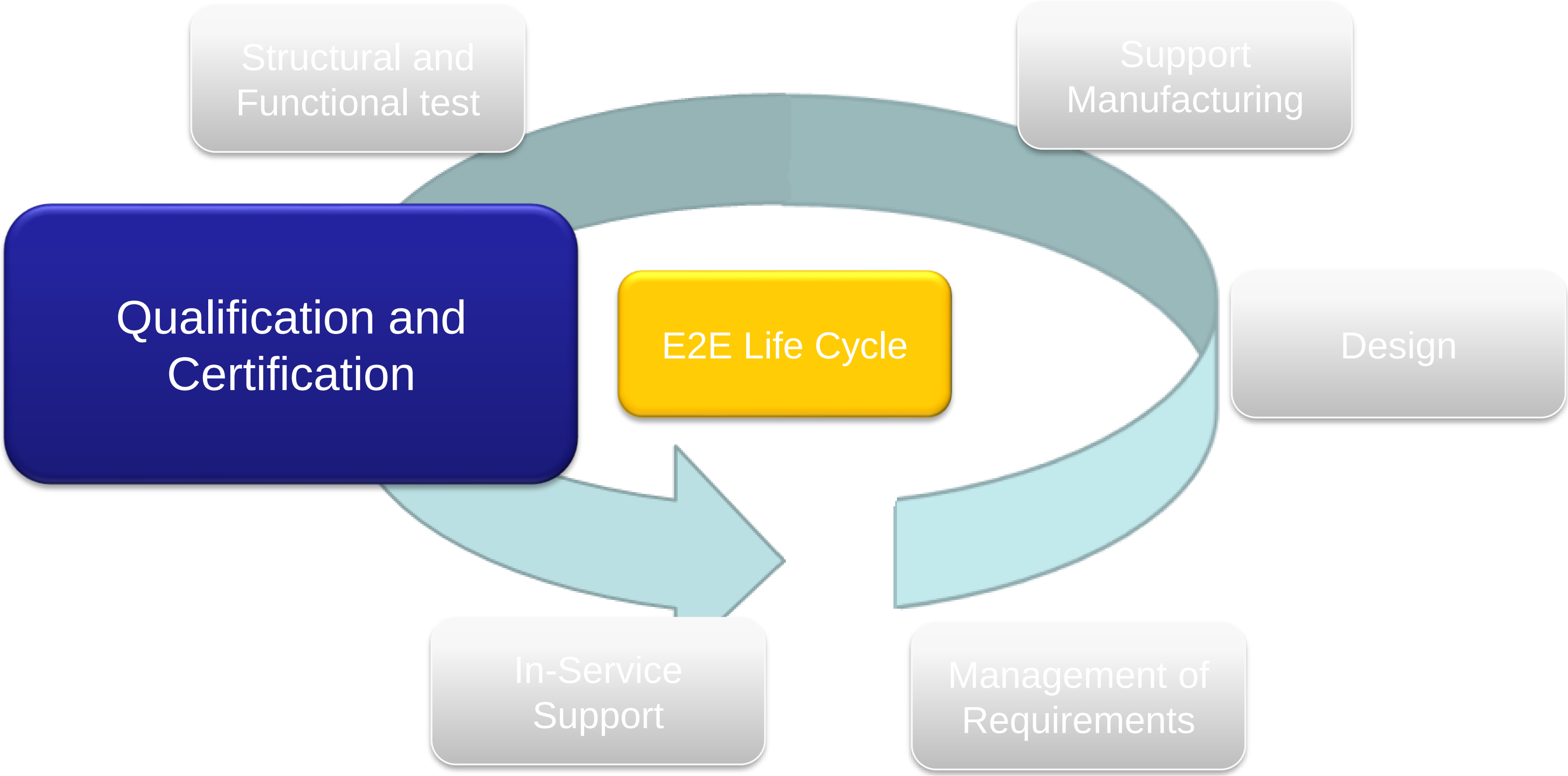






Structural and Functional test

E2E life cycle



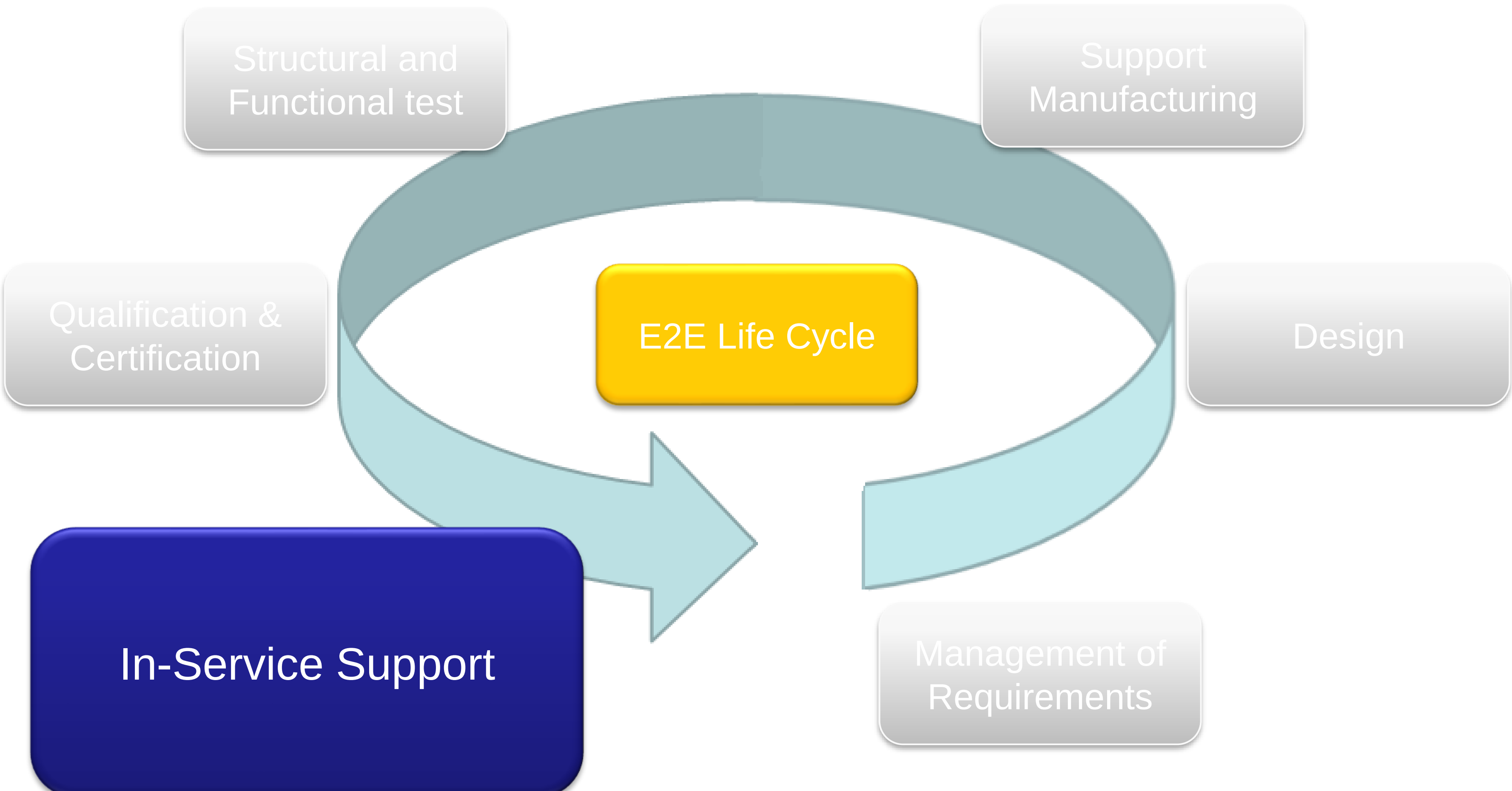
Qualification and Certification

Main activities in Structural Design

- Qualification.
  - Perform validation and verification activities at equipment, component level for qualification.
  - Preparation and Approval of the Technical Specification.
  - Support procurement during bidding Process.
  - Qualification Test (refer to Structural & Functional Test)

- Certification .
  - Certification meetings with the aviation authorities.
  - Preparation of Certification documents.

E2E life cycle



Main activities in Structural Design

- Prepare and Approval of Service Bulletin (SSBB).
- Definition of in-service repairs.
- Provide the data design source of the SRM.
- Resolution of the MISI.
- Resolution of the in-service issues identified by the P&FS committee.

Link to Practical Exercise

Concurrent Engineering.

The product development has to be done with the Concurrent engineering processes with:

- extensive use of the Aircraft Digital Mock Up (DMU)
- use of common harmonized tools, to share all the data's through the extended enterprise
- mature and deployed tools
- teams training performed in due time.

The use of Plateau's approach, to get the best efficiency and communication between teams, is a common practice to fit with targeted schedule. This implies:

- make available work space
- consider resources ramp-up not forgetting growth potential (anticipation of ramp-up and places for temporary additional working group on plateau's).

The needed resources, with required skills, trained on Program applicable tools, are required in due time. The associated Work Stations, in line with program tools, are also mandatory for each skill (designers, stress men…).

L01. References

1. Michael Chun Niu – Airframe Structural Design
2. Structure design principles
3. Design manual